school mathematics ability that is observable in nationally standardized tests of scholastic achievement.

The identified modest yet stable relationship between our intuitive sense of number (i.e., the ANS) and our more formal mathematical abilities across the age span (i.e., 11–85 y) helps to motivate an investigation into the developmental changes in these abilities during these years. One possibility, untested by the present study, is that the ANS may serve as a foundational intuitive workspace for building confidence concerning mathematical transformations, a confidence that may carry over throughout more formal mathematical learning.

Acquiring large heterogenous samples via the internet creates the opportunity to investigate developmental change across the lifespan. The present data reveal changes in *w*, RT, and self-reported mathematics ability across ages. Population trends reveal three dominant transitions in ANS precision (*w*, RT) across the age span with a rapid decrease in RT from 11 to 16 y of age, continued improvements in *w* from 16 to 30 y of age, and a sustained age-related decline in ANS precision (*w*, RT) from 30 to 85 y of age (Fig. 3). The initially high *w* scores at younger ages indicate noisier ANS representations that are gradually refined during the school-age years, with optimal precision (e.g., lowest *w*) attained surprisingly late in life at ~30 y of age (Fig. 3). Most other cognitive abilities, with the notable exception of face recognition ability (29), appear to peak earlier. This extends previous studies, which suggested developmental changes in *w* in smaller laboratory-based samples (9, 10). The large overlap among interdecile ranges across the lifespan (Fig. 3, black bars) indicates that, even after developmental improvements, individual differences in *w* and RT remain large. For example, one adult in eight has a number sense that is less precise (i.e., a larger *w* and slower RT) than a typical 11-y-old child's, indicating that a wide range of ages may stand to benefit from interventions designed to improve ANS precision.

Developmental change across the lifespan is more easily experienced in dynamic graphs in which age is displayed across time. These developmental transitions are illustrated in the interactive scatterplot in Movie S1.

Success in school mathematics, and our performance in mathematics throughout our adult lives, emerges from many factors. The present findings suggest that one such cognitive component is the precision of the ANS, but this in no way implies that a person's precision is immutable or is determined from birth. Indeed, the encouragingly protracted course of developmental change in ANS precision and the large individual differences across the lifespan raise the possibility of interventions to improve number sense across a range of ages.

Understanding how mental abilities change over the course of an entire life is a formidable challenge. One important approach has been to examine change longitudinally, following an individual person through maturation and changes in experience. However, the labor required for such studies is nearly prohibitive if one wishes to characterize change over the course of a full >70-y lifetime. Measuring abilities in tens of thousands of people across various ages throughout development is a complementary approach, and access to the enormous and varied population that uses the Internet holds great promise. Here, we measured the precision of the number sense in more than 10,000 individuals and present a picture of dramatic changes in this system of core cognition and its modest but highly stable and specific relationship with self-reported school mathematics performance across the lifespan.

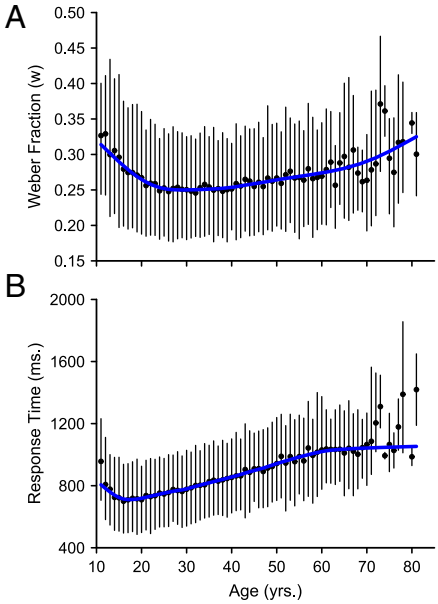


Fig. 3. Developmental changes. (*A* and *B*) *w* and RT (in ms) population trends computed as weighted cubic spline fit to *w* and RT means (blue lines) and interdecile ranges (10th to 90th; black vertical bars).

Methods

Visitors freely navigated to the Web site www.testmybrain.org to participate in various psychological tests and chose the ANS number task. After self-reported school abilities were assessed by questionnaire, ANS precision was assessed with a version of the ANS dots test (current versions available at http://panamath.org/). In the versions of the ANS test presented here, participants were shown brief displays (200 ms) of intermixed blue and yellow dots with five to 20 dots per color and judged whether there were more blue dots or more yellow dots in each display. Stimuli were generated by using parameters described elsewhere (23). Our Web interface allowed us to record accuracy and RT on each trial within an error of ±15 ms. Individual trials with RTs longer than 4 s were excluded to control for inattention. In experiment 1, 10,590 unique participants completed the questionnaire and ANS dots test, 42 of whom were excluded from our final sample because their *w* deviated from the population mean by more than four SDs (*w* and RT scores were roughly normally distributed; Fig. S2). All results remain similar and significant if these subjects are retained, but *w* scores can run off toward infinity with poor performance and likely do not reflect true abilities (e.g., random guessing). In experiment 2, 3,033 unique participants completed the questionnaire and ANS dots test, 27 of whom were excluded from our final sample because their *w* deviated from the population mean by more than four SDs. Percent correct on the ANS test as a function of ratio was modeled for each individual subject as 1 minus error rate, with error rate defined in Eq. **1** in which "erfc" represents the complimentary error function:

$$= \frac{1}{2}\mathrm{erfc}\left(\frac{n_1 - n_2}{\sqrt{2}w\sqrt{n_1^2 + n_2^2}}\right) \qquad [1]$$

This model determines the amount of error in the underlying approximate number representations for the two sets displayed on a trial (n_1 and n_2, i.e., blue dots and yellow dots), specified by a single free parameter, *w*. Correlations presented were between this estimate of ANS precision (i.e., *w*), average RT, self-reported school abilities, and scholastic achievement scores (i.e., SAT).

Concerning our use of a Web-based sample, data collected from Web-based samples have been found to be trustworthy estimates of true abilities, given appropriate subject filtering methods as were used in the present study (26–28). Data collected from www.testmybrain.org, in particular, have been found to be highly reliable (30); to yield laboratory-replicated developmental, aging, and individual differences effects (29, 30, 38, 39); and to be comparable to data gathered in traditional laboratory-based samples on performance-related metrics (30, 39).

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